Numerical Analysis

Assignment #5

Please hand in the C or Matlab code (.m files), graphics, and a brief description of your reasoning as well as comments if any.

Please upload to the 'hw5' directory if you submit your homework in time.

Problem 1:

Use the most accurate three-point formula to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
1.1	9.025013	
1.2	11.02318	
1.3	13.46374	
1.4	16.44465	

b.

x	$f(x)$	$f'(x)$
8.1	16.94410	
8.3	17.56492	
8.5	18.19056	
8.7	18.82091	

代码输出：

--- Table a ---

Results for x values:

f'(1.1) = 17.142505

f'(1.2) = 22.193635

f'(1.3) = 27.107350

f'(1.4) = 32.709255

--- Table b ---

Results for x values:

f'(8.1) = 3.076600

f'(8.3) = 3.116150

f'(8.5) = 3.139975

f'(8.7) = 3.153425

Problem 2：

Suppose that $N(h)$ is an approximation to M for every $h > 0$ and that

$$M = N(h) + K_1h^2 + K_2h^4 + K_3h^6 + \cdots,$$

for some constants $K_1, K_2, K_3, \ldots$. Use the values $N(h)$, $N\left(\frac{h}{3}\right)$, and $N\left(\frac{h}{9}\right)$ to produce an $O(h^6)$ approximation to M.

第一步：消除 $O(h^2)$ 项，得到 $O(h^4)$ 的近似

(1) $M = N(h) + K_1h^2 + K_2h^4 + O(h^6)$

(2) $M = N(h/3) + K_1(h/3)^2 + K_2(h/3)^4 + O(h^6) = N(h/3) + (K_1/9)h^2 + (K_2/81)h^4 + O(h^6)$

(2) ×9 - (1):

$9M - M = [9N(h/3) - N(h)] + [(K_2/9) - K_2]h^4 + O(h^6)$

$8M = [9N(h/3) - N(h)] - (8/9)K_2h^4 + O(h^6)$

定义 $N_1(h) = [9N(h/3) - N(h)] / 8$

$M = N_1(h) - (K_2/9)h^4 + O(h^6)$。这是一个 $O(h^4)$ 的近似。

第二步：消除 $O(h^4)$ 项，得到 $O(h^6)$ 的近似

$M = N_1(h) + K'_2h^4 + O(h^6)$，其中 $K'_2 = -K_2/9$。

用 h/3 替换 h，得 $M = N_1(h/3) + K'_2(h/3)^4 + O(h^6) = N_1(h/3) + (K'_2/81)h^4 + O(h^6)$。

现在有两个 $O(h^4)$ 的方程:

(3) $M = N_1(h) + K'_2h^4 + O(h^6)$

(4) $M = N_1(h/3) + (K'_2/81)h^4 + O(h^6)$

(4) ×81 - (3):

$81M - M = [81N_1(h/3) - N_1(h)] + O(h^6)$

$80M = 81N_1(h/3) - N_1(h) + O(h^6)$

$M = [81N_1(h/3) - N_1(h)] / 80$。这就是 $O(h^6)$ 的最终近似 $N_2(h)$。

第三步：代入 N_1 的表达式

$N_1(h) = [9N(h/3) - N(h)] / 8$

$N_1(h/3) = [9N(h/9) - N(h/3)] / 8$

$M \approx N_2(h) = (1/80) * [81 * ((9N(h/9) - N(h/3)) / 8) - ((9N(h/3) - N(h)) / 8)]$

$M \approx (1 / (80*8)) * [81(9N(h/9) - N(h/3)) - (9N(h/3) - N(h))]$

$M \approx (1 / 640) * [729N(h/9) - 81N(h/3) - 9N(h/3) + N(h)]$

$M \approx (1 / 640) * [729N(h/9) - 90N(h/3) + N(h)]$

故 $O(h^6)$ 的近似值由下式给出：

$M \approx [729 * N(h/9) - 90 * N(h/3) + N(h)] / 640$

Problem 3:

Approximate the following integrals using the Trapezoidal rule and Simpson's rule, respectively.

a. $\int_{-0.25}^{0.25} (\cos x)^2 \, dx$

b. $\int_{-0.5}^{0} x \ln(x+1) \, dx$

c. $\int_{0.75}^{1.3} \left((\sin x)^2 - 2x \sin x + 1\right) \, dx$

d. $\int_{e}^{e+1} \frac{1}{x \ln x} \, dx$

代码输出：

--- Approximations for Integrals ---

Integral (a):
  Trapezoidal Rule: 0.469149
  Simpson's Rule: 0.479426

Integral (b):
  Trapezoidal Rule: 0.086643
  Simpson's Rule: 0.084411

Integral (c):

Trapezoidal Rule: 0.043810
Simpson's Rule: 0.043232

Integral (d):
Trapezoidal Rule: 0.320503
Simpson's Rule: 0.321481

Problem 4:

Use Romberg integration to compute $R_{3,3}$ for the following integrals.

a. $\int_{-1}^{1} (\cos x)^2 \, dx$

b. $\int_{-0.75}^{0.75} x \ln(x+1) \, dx$

c. $\int_{1}^{4} \left((\sin x)^2 - 2x \sin x + 1\right) \, dx$

d. $\int_{e}^{2e} \frac{1}{x \ln x} \, dx$

代码输出：

--- Romberg Integration Results ---
Integral (a): R_3,3 = 0.909337
Integral (b): R_3,3 = 0.000000
Integral (c): R_3,3 = 0.693489
Integral (d): R_3,3 = 0.405465

Problem 5 (Could be done after Tuesday's course):

Use Euler's method to approximate the solutions for each of the following initial-value problems.

a. $y' = y/t - (y/t)^2, \quad 1 \le t \le 2, \quad y(1) = 1, \text{ with } h = 0.1$

b. $y' = 1 + y/t + (y/t)^2, \quad 1 \le t \le 3, \quad y(1) = 0, \text{ with } h = 0.2$

代码输出：

--- Euler's Method Results for IVP (a) ---

t_i	w_i
1.00	1.000000
1.10	1.000000
1.20	1.008264
1.30	1.022641
1.40	1.041352
1.50	1.063184
1.60	1.087266
1.70	1.112959
1.80	1.139803
1.90	1.167445
2.00	1.195612

--- Euler's Method Results for IVP (b) ---

t_i	w_i

1.00	0.000000
1.20	0.200000
1.40	0.438889
1.60	0.720494
1.80	1.043513
2.00	1.402685
2.20	1.791985
2.40	2.206103
2.60	2.640656
2.80	3.092193
3.00	3.557945